
QUARRYING AS AN ENVIRONMENTAL ETHIC CRISIS

MARK GREGORY G. HONTIVEROS
College of Law, West Visayas State University

History

Stone extraction for construction materials is the primary use of quarries. The use of quarries dates back thousands of years. The Great Pyramids were constructed by the inhabitants of Kemet, ancient Egypt, using enormous hand-cut limestone and granite stones from surrounding quarries.¹ These blocks weigh many tons each. In marble, granite, and limestone quarries in ancient Rome, the incredibly hard task of cutting stones was frequently assigned to enslaved and imprisoned individuals.

The techniques for removing stone and other commodities from quarries have evolved since the first mines were established in the Egyptian region of Aswan.² Hammers, picks, and chisels made of stone or metals like iron and bronze were used to mine the first quarries. It took a lot more work to quarry material for use in building materials. Stones had to be hand-hauled or transported out of quarries.³ Moreover, pulley systems with ropes and movable wooden sleds and tracks might be used to lift stones.

Definition

A quarry is a place where rocks, sand, or minerals are extracted from the surface of the Earth.⁴ A quarry is a type of mine called an open-pit mine because it is open to Earth's surface. Another type of mine, a subsurface mine, consists of underground tunnels or shafts. Additionally, asphalt and concrete are made with aggregate.⁵ Because of this, concrete and asphalt plants are frequently constructed next to quarries. When building roads, aggregate is combined with an oily material called asphalt. The ancient Romans created concrete, which is a combination of aggregate and sticky stone cement. The extensive network of roads and aqueducts constructed by the Romans, much of which is still in use today, was made possible by concrete and aggregate.

¹ Rutledge, et. al., National Geographic, (October 19, 2023), <https://education.nationalgeographic.org/resource/quarry/>

² *Id.*

³ *Id.*

⁴ *Id.*

⁵ *Id.*

The Constitution

Article II, Section 16 of the Philippine Constitution provides that:

“The State shall protect and advance the right of the people to a balanced and healthful ecology in accord with the rhythm and harmony of nature.”⁶

Policy of the State

Section 2 of Presidential Decree No. 1151 provides that:

“it shall be the responsibility of the Government, in cooperation with concerned private organizations and entities, to use all practicable means, consistent with other essential considerations of national policy, in promoting the general welfare to the end that the Nation may (a) recognize, discharge and fulfill the responsibilities of each generation as trustee and guardian of the environment for succeeding generations, (b) assure the people of a safe, decent, healthful, productive and aesthetic environment, (c) encourage the widest exploitation of the environment without degrading it, or endangering human life, health and safety or creating conditions adverse to agriculture, commerce, and industry, (d) preserve important historic and cultural aspects of the Philippine heritage, (e) attain a rational and orderly balance between population and resource use, and (f) improve the utilization of renewable and non-renewable resources.”⁷

The same Decree likewise provides that:

“It shall be the duty and responsibility of each individual to contribute to the preservation and enhancement of the Philippine environment.”⁸

While the Philippines needs to strengthen its law enforcement capacity, it will also benefit from building on its rich well-crafted environmental policies with a new model of environmental governance focused on well-being and guided by principles of sustainability, ecosystem health, precaution, interconnectedness, and inclusiveness.

An Environmental Concern

For an extended period, environmental issues have been escalating and widespread worldwide, particularly in the Philippines. The Philippines is renowned for both its breathtaking scenery and its wealth of natural resources, which include stunning mountains, unspoiled islands, riverbanks, and other features that offer vital

⁶ CONST. (1987) art. II, §16 (Phil.).

⁷ Philippine Environmental Policy, Pres. Dec. 1151 (June 06, 1977) (Phil),

⁸ *Id.*

ecosystem services to the local population.⁹ It is recognized for having a distinct and abundant biodiversity, where uncommon flora and animals can be found around the entire country. But because of its many non-eco-friendly activities that worsen the environment and cause typhoons, landslides, and other natural disasters, the Philippines is also one of the many countries that regularly experience these natural disasters.¹⁰ It is also affected by climate change. Quarrying is one of the most destructive man-made activities commonly carried out in the Philippines. The process of quarrying involves taking stones or building materials out of the ground and along riverbanks.¹¹

There are different kinds of quarrying: river and mountain quarrying.¹² They both have a significant detrimental impact on the environment. Issues surrounding quarrying have been covered by the media and are still ongoing. Indeed, reports of illicit quarrying operations have surfaced nationwide, operating on Philippine beaches, riverbanks, and mountains in particular regions and rural areas. These operations have resulted in coastal erosion, flooding in coastal communities, and even accidents and fatalities.

Mountain quarrying entails removing highland soil and plants, destroying the value of the removed region. In contrast, river quarrying is the most common type of quarrying in the Philippines, and it extracts sand and gravel. River quarrying converts riverbeds into big and deep holes, lowering the groundwater table and causing drinking water wells atop the river's embankments to dry up. In coastal areas, saltwater water may seep into freshwater bodies. Aside from that, quarrying reduces soil cohesiveness, widens riverbanks, and lowers their height. It also causes uneven riverbed deepening, which can lead to riverbank degradation and environmental damage.¹³

There is a greater potential for pollution, soil erosion, floods, and siltation with these types of quarry operations. In addition, the transportation of these quarrying

⁹ De Burca, J. (March 21, 2024), <https://constructive-voices.com/philippines-biodiversity-and-the-built-environment-2/>

¹⁰ World Bank Group., Towards a Comprehensive Disaster Risk Management System for the Philippines (March 07, 2023), <https://www.worldbank.org/en/country/philippines/brief/towards-a-comprehensive-disaster-risk-management-system-for-the-philippines>

¹¹ The Rise of Environmental Sustainability in the Philippines (April 24, 2023), <https://standard-insights.com/blog/rise-of-environmental-sustainability-in-the-philippines/>

¹² Reyes, M.A., Bad Eggs (September 24, 2023), <https://www.pids.gov.ph/details/news/in-the-news/bad-eggs>

¹³ Das et.al., Riverbank Erosion (2014), <https://www.preventionweb.net/understanding-disaster-risk/terminology/hips/gh0028>

activities and the noise pollution in the quarry locations produce dust.¹⁴ Furthermore, throughout the aggregate extraction and processing phases, engineering activities directly contribute to a portion of the environmental disturbances caused by quarrying. Even though there are some short-term effects of quarrying that are simple to predict and monitor, it is still preferable to remain vigilant and avoid complacency.

Currently, it is undeniable that quarrying plays a vital role in society by creating necessary structures like roads, schools, bridges, and buildings, thus aiding in the overall development of the country. It also creates employment opportunities and plays a significant role in the growth of the contemporary economy.

Nevertheless, it is also noted that mining operations pose a danger to the daily lives of people. The World Health Organization states that substandard environmental conditions are a significant factor in poor health and quality of life, impairing sustainable development, and emphasizing the vital role of the environment in human health. Everyday life may be affected by some of the adverse consequences of quarrying.¹⁵

Air Pollution from Quarrying

Dust produced by quarries is a significant contributor to air pollution, with the impact varying based on factors such as local microclimate conditions, dust concentration in the air, particle size, and chemical composition. For instance, limestone quarries generate alkaline and reactive dust, while coal mines produce acidic dust.¹⁶

Air pollution is not only bothersome as it can affect surfaces and health, especially for those with respiratory issues, but also harms plants by blocking and damaging internal structures, causing abrasion on leaves and cuticles, and potentially affecting long-term survival with chemical effects.¹⁷

¹⁴ Worku, H., Environmental and socioeconomic impacts of cobblestone quarries in Addis Ababa and implication for resource use efficiency, environmental quality, and sustainability of land after-use (March 02, 2018), <https://onlinelibrary.wiley.com/doi/10.1002/tqem.21524>

¹⁵ Health and the Environment (n.d.) World Health Organization, <https://www.who.int/westernpacific/about/how-we-work/programmes/health-and-the-environment>

¹⁶ The Impact of Quarrying: Cause and Effect (January 7, 2024), <https://organiconaturalpaint.co.uk/blogs/the-impact-of-quarrying-cause-and-effect/>

¹⁷ *Id.*

Apart from dust pollution, quarrying also generates carbon emissions from heavy machinery, material transportation, and energy-intensive stone processing. These actions emit greenhouse gasses, which add to the problem of climate change.

Noise Pollution from quarrying

Regrettably, mining includes multiple activities that produce substantial levels of noise. The process begins with preliminary tasks like building roads or railway access, setting up a compound, and constructing mineral processing plants. Afterward comes the stage of revealing the mineral to be taken out, typically accomplished by clearing away the surface soil and other layers with equipment like scrapers hydraulic excavators, and dump trucks. The extraction of the mineral will create significant noise, especially when blasting techniques are utilized. Afterward, the utilization of motorized equipment to move the materials and the potential installation of factories to crush and sort the minerals, both add to the noise pollution in the surroundings.¹⁸

Damage to Biodiversity from quarrying

One major downside of quarrying on the environment is the harm it causes to biodiversity. Biodiversity mainly involves a variety of living organisms, such as fish, insects, invertebrates, reptiles, birds, mammals, plants, fungi, and microorganisms. Preserving biodiversity is crucial because all species are connected, even if not obvious or acknowledged, and our survival relies on the delicate equilibrium in nature.

Quarrying can devastate habitats and the wildlife they sustain. While excavations may not directly eliminate habitats, they can still cause harm by indirectly impacting the environment, leading to changes in groundwater or surface water levels that could result in habitats drying up or becoming flooded. Noise pollution can profoundly impact certain species and hinder their ability to reproduce successfully. However, by carefully planning and managing, it is feasible to reduce the impact on biodiversity. Additionally, quarries can offer a chance to establish new habitats or restore current ones.¹⁹

Additionally, mining in riverbeds mainly impacts the environment, the community, and the wildlife. The destruction of both public and private properties can

¹⁸ The Impact of Quarrying: Cause and Effect (January 7, 2024), <https://organicnaturalpaint.co.uk/blogs/the-impact-of-quarrying-cause-and-effect/>

¹⁹ *Id.*

harm the lives of residents living nearby, particularly in rural areas where people rely heavily on the resources of their surroundings. Additionally, it has the potential to result in fatalities and serious incidents by triggering deadly landslides. Quarrying activities can cause significant harm to the environment, such as the destruction of plants, erosion of soil, siltation of rivers, pollution from dust, and landslides. In 2018, it was reported that quarrying operations in seven regions in the Philippines were temporarily closed because an island in Cebu City close to a limestone quarry was affected by a landslide triggered by typhoon Mangkhut which killed approximately twenty-nine (29) people and some were trapped under the debris. This alone served as a warning that quarry operation is indeed deadly and harmful.²⁰

While not everyone may be aware of this environmental problem, quarrying is widely recognized as a highly hazardous activity that impacts the lives of numerous individuals. This activity leaves significant long-term economic, social, and environmental marks.²¹ The rise in quarrying activities is linked to various social challenges such as health and safety risks, displacement of communities, destruction of cultural sites, and utilization of villages for quarries.

How has the issue arisen?

Quarrying operations in the Philippines have become vital for economic growth and infrastructure development due to increasing demands from development, population growth, and construction needs. However, this activity has also raised concerns about its impact on the environment.²²

This problem is now a major concern and quite worrying as illegal quarrying continues in the Philippines despite the presence and enforcement of environmental laws, due to inadequate supervision by authorities, strict regulations on quarry rehabilitation, and the absence of post-quarrying programs to mitigate its negative

²⁰ Marquez B., Philippines: At least 29 dead in Cebu landslide. Aljazeera. (September 21, 2018), <https://www.aljazeera.com/news/2018/9/21/philippines-at-least-29-dead-in-cebu-landslid.e>

²¹ Lad R.J. and Samant J.S. Environmental And Social Impacts Of Stone Quarrying- A Case Study Of Kolhapur District (March 25, 2014), https://www.academia.edu/6548627/ENVIRONMENTAL_AND_SOCIAL_IMPACTS_OF_STONE_QUARRYING_A_CASE_STUDY_OF_KOLHAPUR_DISTRICT

²² Tamayo, P., Characterizing the Effects of Quarrying Industry in Northern Philippines: A Mixed-methods Study. (2023), https://www.researchgate.net/publication/371342275_Characterizing_the_Effects_of_Quarrying_Industry_in_Northern_Philippines_A_Mixed-methods_Study

effects on the environment.²³ Additionally, many developing countries have neglected to rehabilitate most quarrying sites after the cessation of quarrying activities. Should these issues and the mining activities continue without control, there is no question that this ecological issue will escalate into a more severe threat to the lives of Filipinos.

Political Factor

The government plays various roles in environmental issues. Their responsibility is to safeguard the environment and its resources, ensuring that environmental activities are sustainable and benefit everyone equally. They also work to protect the properties of those living near quarry sites, particularly in indigenous communities. Politics plays a role in shaping environmental laws, and there is a contentious debate about the potential political influence on quarrying activities that have a detrimental impact on the environment. Certain government departments neglected to monitor quarry operations and did not conduct proper reviews of quarry owners' permits, allowing illegal quarrying to persist and thrive. Companies must still obtain permits to legally extract minerals from the city when quarrying. Getting permits approved and signed by the government is necessary, but it often leads to complications.

The main task in mining operations is transporting materials using dump trucks.²⁴ This is a daily operation that drivers repeat through the same routes, making it a tedious and challenging task for them. Dump trucks utilized in quarries require a significant investment, making it crucial to prioritize road maintenance to boost transportation efficiency, decrease maintenance expenses, and extend the lifespan of tires and machinery overall. Road maintenance involves flattening gravel roads, but quarry road conditions change daily due to mining activities. As stone extraction operations continue, new roads are built and existing roads are altered, making current maps outdated.

²³ Department of Environment and Natural Resources (DENR) (July 2, 2021), <https://denr.gov.ph/news-events/denr-enforcement-operatives-swoop-down-on-illegal-quarrying/>

²⁴ Types of Mining and Their Respective Equipment (n.d.), <https://www.cashmanequipment.com/about/the-dirt-blog/common-types-of-mining-equipment-used>

Issues in the Government

In 2024, the Provincial Government of Camarines Norte investigated reports of illicit quarry activities in different locations throughout the province.²⁵ A contractor reportedly engaged in a river control project in Barangay Bakiad and Sta is said to have conducted unauthorized excavation along the shores. Even though authorities carried out a joint inspection, no individuals were arrested at the location. The Inter-Agency Task Force on Environmental Protection has shown dissatisfaction regarding the discovery.²⁶ As stated by the Department of Public Works and Highways (DPWH), it is strictly against the rules to conduct any extraction or quarrying activities within 10 meters of government buildings or one kilometer of a bridge.

In 2018, Iloilo mulls mountain quarrying amid a shortage of sand, and gravel. The use of mountain soil as an alternative to address the shortage of sand and gravel for the construction of various projects implemented by the DPWH in Iloilo is being explored. The Iloilo government considers mining in the mountains due to a lack of sand and gravel. Exploring the use of mountain soil as a substitute for sand and gravel in DPWH projects in Iloilo to combat shortages.²⁷

The Regional Director of DPWH-6 stated that there were approximately PHP11 billion worth of projects allocated for Iloilo province in 2018. The budget is higher than the PHP8 billion allocated for projects last year. He mentioned that the projects in the province for that year required 329,621 cubic meters of sand and 537,722 cubic meters of gravel, including roads, bridges, and schools. He also mentioned that he asked other provincial governors in the area to give priority to contractors when granting quarry permits.²⁸

In the same way, in 2014, DPWH Region 11 called on the Davao City government to prevent illegal quarrying near bridges. DPWH 11 regional director Mariano Alquiza said that the degradation of the bridge's foundation is largely due to illegal quarrying. Mayor Rodrigo R. Duterte of Davao City released Executive Order (EO) No. 40, also known as "An Act Establishing the Task Force Subasa Dakbayan (TF Suba)" to address environmental issues in line with current laws protecting the river. The command indicates that the extensive unauthorized quarry activities along

²⁵ GMA Network., Alleged illegal quarrying in Camarines Norte looked into (November 24, 2023), <https://www.gmanetwork.com/regionaltv/news/99638/alleged-illegal-quarrying-in-camarines-norte-looked-into/story/>

²⁶ *Id.*

²⁷ Panay News, Local Newspaper, Iloilo running out of sand, gravel (November 10, 2016), <https://www.panaynews.net/iloilo-running-out-of-sand-gravel/>

²⁸ Ferrer, C., Iloilo mulls mountain quarrying amid shortage of sand, gravel, Philippine News Agency (July 12, 2018), <https://www.pna.gov.ph/articles/1041295>

the primary river in the urban area have been observed and are expected to pose a threat not just to the people living nearby but also to the safeguarding and maintenance of the rivers.²⁹

Why is quarrying even needed?

Quarries, also referred to as open-cast mines,³⁰ are locations where raw mineral material is removed. This is crucial for numerous secondary products and industries, which are essential for achieving a clean, sustainable net-zero society. All things surrounding us are composed of either organic (biologically derived) or inorganic (mineral-derived) materials. The majority of products consist of a combination of both.

The demand for specific metallic minerals like lithium, selenium, and indium is increasing rapidly due to the production of EVs, wind turbines, and solar panels.³¹ Mined minerals are used to make high-performance glazing glass, insulating mineral wool, and other eco-friendly building materials, along with natural fertilizers that substitute petrochemical-derived ammonium nitrates.

Quarry Location

Shortening the distance from the quarry to the customer can decrease the traffic and emissions linked to transportation. Yet, when the site is close to a village or another populated area, inhabitants may be concerned about the potential increase in traffic, noise, and dust from a new quarry. If not controlled, wildlife can be greatly affected by silt runoff, reduced water table levels, and habitat loss.

On a worldwide scale, when minerals are imported from countries with lower costs, it often means transferring problems to countries with less regulation and enforcement of environmental, social, and ethical protections. In certain nations, this has affected not just animals, but also led to the relocation of individuals. Irregular and illegal mining.³²

²⁹ Edge Davao., DPWH asks city gov't to keep quarrying away from bridges (October 2, 2014), <https://edgedavao.net/the-big-news/2014/10/dpwh-asks-city-govt-to-keep-quarrying-away-from-bridges/>

³⁰ Atlantic Pumps, Quarrying & Mining – Good or Bad for the Environment? (January 11, 2024), <https://atlanticpumps.co.uk/blogs/news/quarrying-mining-good-or-bad-for-the-environment>

³¹ *Id.*

³² Atlantic Pumps, Quarrying & Mining – Good or Bad for the Environment? (January 11, 2024), <https://atlanticpumps.co.uk/blogs/news/quarrying-mining-good-or-bad-for-the-environment>

Government Approach

Recognizing the significance of our waterways, DPWH released Department Order No. 139, Series of 2014³³ to establish regulations for river dredging in flood-control projects and quarrying activities. To prevent harm to the environment caused by quarrying, project sites must undergo a thorough soil investigation, which involves conducting on-site bathymetric, topographic, and hydrographic surveys in line with the department's Design Guidelines Criteria and Standards.

A thorough investigation and analysis of the soil must be conducted to avoid alterations in the morphology of the stream. It is necessary to obtain the appropriate environmental permits, such as the Certificate of Non-Coverage (CNC) or Environmental Compliance Certificate (ECC).³⁴ An exhaustive plan and design report for dredging must also be presented to the DPWH offices as outlined by DENR's delegated authorities. Moreover, before the approval of projects, they must adhere to the current laws and regulations of the Water Code of the Philippines.³⁵ and the Philippine Mining Act of 1995,³⁶ among other requirements.

The DPWH Project Management Office (PMO), along with all regional and district offices, must follow instructions to protect waterways' morphology during flood-control projects, dredging, and quarrying operations nationwide.

Local officials are advised to follow DENR and DPWH recommendations before authorizing quarrying activities near rivers and water bodies.³⁷

Philippine Jurisprudence

In the case of the *Province of Pampanga v. Executive Secretary Romulo*,³⁸ Associate Justice Lazaro-Javier, in her concurring opinion, said that quarry operations

³³ DPWH Issues Guidelines for River Dredging Operations, (December 23, 2014), <https://www.dpwh.gov.ph/dpwh/news/1135>

³⁴ DPWH Issues Guidelines for River Dredging Operations, (December 23, 2014), <https://www.dpwh.gov.ph/dpwh/news/1135>

³⁵ A Decree Instituting a Water Code, Thereby Revising and Consolidating the Laws Governing the Ownership, Appropriation, Utilization, Exploitation, Development, Conservation and Protection of Water Resources, Pres. Dec. 1067, s. 1976

³⁶ An Act Instituting a New System of Mineral Resources Exploration, Development, Utilization, and Conservation, Rep. Act 7942, (March 3, 1995) (Phil.).

³⁷ DPWH Issues Guidelines for River Dredging Operations (December 23, 2014), <https://www.dpwh.gov.ph/dpwh/news/1135>

³⁸ *Province of Pampanga v. Executive Secretary Romulo*, G.R. No. 195987, (January 12, 2021) (Phil.).

do not involve massive undertakings. Quarry operations may likewise be done utilizing small-scale equipment and methods. Quarrying and small-scale mining are not mutually exclusive terms. Small-scale mining may include quarry operations, though quarry operations cannot extend to small-scale mining.³⁹ The description of quarrying as the extraction by light blasting and barring down with steel bars and crushing by manual labor and crushers is attuned with R.A. No. 7076's definition of small-scale mining, i.e., heavy reliance on manual labor using simple implements and methods, without using explosives or heavy mining equipment.

Also, the Supreme Court has issued a writ of kalikasan against the Department of Environment and Natural Resources (DENR), Mines and Geosciences Bureau (MGB), Ipilan Nickel Corporation (INC), and Celestial Nickel Mining and Exploration Corporation (Celestial Mining) ordering them to file a verified return of the writ within a non-extendible period of 10 days from receipt of the order.⁴⁰ On May 30, 2022, the DENR Community Environment and Natural Resources Office issued a cease-and-desist order⁴¹ directing INC to halt its port construction for the lack of a valid Miscellaneous Lease Agreement approved by the DENR MIMAROPA Regional Office. This was followed by a resolution by the Sangguniang Bayan of Brooke's Point requesting the MGB to investigate the mining areas claimed by INC and recall issued permits pending investigation. However, no action was taken by the MGB on the request. This prompted the Sangguniang Bayan to issue another resolution urging President Ferdinand R. Marcos, Jr., through the DENR Secretary, to investigate INC's operation despite the absence of the following crucial government permits: (1) clearance from the Protected Area Management Boards; and (2) Certificate of Precondition from NCIP, as well as the extension of the MPSA without the endorsement of the various local government units of Brooke's Point.

The Court also applied the precautionary principle, as laid down in the 2021 case of *PTK2 H2O Corp. v. Court of Appeals*.⁴² Under the principle, a project proponent is required to provide evidence to dispel concerns regarding the potentially harmful impact of a project on the environment, shifting the burden of evidence of harm away from those likely to suffer harm and onto those desiring to change the status quo.

³⁹ *League of Provinces of the Philippines v. Department of Environment and Natural Resources*, 709 Phil. 189, 229-230 (2013).

⁴⁰ *Supreme Court Issues Writ of Kalikasan Against DENR and Mining Operators in Mt. Mantalingahan, Palawan*, Supreme Court of the Philippines, (August 16, 2023), <https://sc.judiciary.gov.ph/supreme-court-issues-writ-of-kalikasan-against-denr-and-mining-operators-in-mt-mantalingahan-palawan/>

⁴¹ Pulta, B., SC issues writ of kalikasan v. mining ops in Palawan, (August 16, 2023), <https://www.pna.gov.ph/articles/1207906>

⁴² *PTK2 H2O Corporation v. Court of Appeals*, 915 Phil. Rep. 119 - 219 (November 16, 2021).

In the present case, the Court found that there is a possibility of serious and irreversible harm to the environment and the inhabitants of Brooke's Point located in the Mt. Mantalingahan Mountain Range. Thus, with the Court's issuance of a writ of kalikasan, the project proponents, INC and Celestial, are required to provide evidence to dispel concerns regarding the potentially harmful impact of a project on the environment.

Who Benefits from Quarrying?

The benefits of quarrying include an increase in the employment rate of the country. According to the Philippine Statistics Authority (PSA), in 2016, with a total of 130 mining and quarrying establishments, there were 28,992 who were employed in this sector. 11.1% of these people or 3,190 of them are employed in the quarrying of stone, sand, and clay. The city also benefits from quarrying. In Davao City, quarrying taxes contribute to the city's annual income by 10%. However, there are those establishments that quarry illegally, leaving the city with a loss of 50% of their potential income. Aside from that, erring government officials who are in great power and the business sectors can benefit from this situation because they can manipulate and control the resources and quarry operations.

Conclusion

It is a significant responsibility to plan and implement a new quarry or mine, as it involves balancing development and preservation needs. Decisions regarding planning and actions taken by quarry operators have the potential to either harm or improve local habitats, the environment, climate regulations, affordable living standards, and employment opportunities in the area.

Engaging in responsible mining and quarrying locally reduces the necessity of transporting important mineral resources worldwide, contributes to a net gain in biodiversity and ecosystems, and benefits the regional economy. Continuous research and innovation are still being conducted to discover novel methods to address current needs while also safeguarding the earth's resources for future generations.

River extraction presents potential harmful effects on the environment. Using heavy machinery like payloaders, backhoes, and dump trucks to push sand and gravel aggregates at the river banks and bed disrupts the soil's natural cohesion. Consequently, the riverbank is ruined and the river's mouth expands. In wet seasons, when there is a high volume of water flowing from upstream, flooding occurs. This leads to the accumulation of sand and gravel at the surface level of vegetation, causing the gradual erosion of topsoil and making it unsuitable for plants. This results in neighboring lands becoming infertile.

Recommendations

A more extensive and collaborative information dissemination effort, with the involvement of academia, Local Government Organizations (LGOs), and Non-Government Organizations (NGOs), is required to raise residents' awareness about the environmental impacts of quarrying and other activities that harm nature.

Laws should be created and enforced to require quarrying contractors to make compulsory contributions to fund environmental projects aimed at restoring the environment damaged by quarrying.

Bibliography

Constitution

1987 Constitution of the Philippines art. II, §16 (Phil.).

Statutes

A Decree Instituting a Water Code, Thereby Revising and Consolidating the Laws Governing the Ownership, Appropriation, Utilization, Exploitation, Development, Conservation and Protection of Water Resources, Pres. Dec. 1067, s. 1976.

An Act Instituting a New System of Mineral Resources Exploration, Development, Utilization, and Conservation, Rep. Act 7942 (March 3, 1995) (Phil.).

Philippine Environmental Policy, Pres. Dec. 1151 (June 06, 1977) (Phil.).

Jurisprudence

League of Provinces of the Philippines v. Department of Environment and Natural Resources, 709 Phil. 189, 229-230 (2013) (Phil.).

PTK2 H2O Corporation v. Court of Appeals, 915 Phil. Rep. 119 - 219 (November 16, 2021) (Phil.).

Province of Pampanga, v. Executive Secretary Romulo. G.R. No. 195987, (January 12, 2021) (Phil.).

Online Sources

Atlantic Pumps., Quarrying & Mining – Good or Bad for the Environment? (January 11, 2024), <https://atlanticpumps.co.uk/blogs/news/quarrying-mining-good-or-bad-for-the-environment>)

Das et.al., Riverbank Erosion (2014), <https://www.preventionweb.net/understanding-disaster-risk/terminology/hips/gh0028>

De Burca, J. (March 21, 2024), <https://constructive-voices.com/philippines-biodiversity-and-the-built-environment-2/>

DPWH Issues Guidelines for River Dredging Operations, (December 23, 2014),
<https://www.dpwh.gov.ph/dpwh/news/1135>

Edge Davao., DPWH asks city gov't to keep quarrying away from bridges (October 2, 2014),
<https://edgedavao.net/the-big-news/2014/10/dpwh-asks-city-govt-to-keep-quarrying-away-from-bridges/>

Ferrer, C. Iloilo mulls mountain quarrying amid shortage of sand, gravel (July 12, 2018). Philippine News Agency,
<https://www.pna.gov.ph/articles/1041295>

GMA Network., Alleged illegal quarrying in Camarines Norte looked into (November 24, 2023),
<https://www.gmanetwork.com/regionaltv/news/99638/alleged-illegal-quarrying-in-camarines-norte-looked-into/story/>

Health and the Environment (n.d.) World Health Organization,
<https://www.who.int/westernpacific/about/how-we-work/programmes/health-and-the-environment>

Lad R.J. and Samant J.S. Environmental And Social Impacts Of Stone Quarrying-A Case Study Of Kolhapur District (March 25, 2014),
https://www.academia.edu/6548627/ENVIRONMENTAL_AND_SOCIAL_IMPACTS_OF_STONE_QUARRYING_A_CASE_STUDY_OF_KOLHAPUR_DISTRICT

Marquez B., Philippines: At least 29 dead in Cebu landslide. Aljazeera. (September 21, 2018),
<https://www.aljazeera.com/news/2018/9/21/philippines-at-least-29-dead-in-cebu-landslide>

Panay News, Local Newspaper. Iloilo running out of sand, gravel (November 10, 2016),
<https://www.panaynews.net/iloilo-running-out-of-sand-gravel/>

Pulta, B., SC issues writ of kalikasan v. mining ops in Palawan, (August 16, 2023),
<https://www.pna.gov.ph/articles/1207906>

Reyes, M.A., Bad Eggs (September 24, 2023),
<https://www.pids.gov.ph/details/news/in-the-news/bad-eggs>

-
- Rutledge, et. al. National Geographic, (October 19, 2023), <https://education.nationalgeographic.org/resource/quarry/>
- Tamayo, P., Characterizing the Effects of Quarrying Industry in Northern Philippines: A Mixed-methods Study. (2023), https://www.researchgate.net/publication/371342275_Characterizing_the_Effects_of_Quarrying_Industry_in_Northern_Philippines_A_Mixed-methods_Study
- The Impact of Quarrying: Cause and Effect (January 7, 2024), <https://organicnaturalpaint.co.uk/blogs/the-impact-of-quarrying-cause-and-effect/>
- The Rise of Environmental Sustainability in the Philippines. (April 24, 2023), <https://standard-insights.com/blog/rise-of-environmental-sustainability-in-the-philippines/>
- Types of Mining and Their Respective Equipment (n.d.), <https://www.cashmanequipment.com/about/the-dirt-blog/common-types-of-mining-equipment-used>
- Worku, H., Environmental and socioeconomic impacts of cobblestone quarries in Addis Ababa and implication for resource use efficiency, environmental quality, and sustainability of land after-use (March 02, 2018), <https://onlinelibrary.wiley.com/doi/10.1002/tqem.21524>
- World Bank Group., Towards a Comprehensive Disaster Risk Management System for the Philippines (March 07, 2023), <https://www.worldbank.org/en/country/philippines/brief/towards-a-comprehensive-disaster-risk-management-system-for-the-philippines>
- Supreme Court Issues Writ of Kalikasan Against DENR and Mining Operators in Mt. Mantalingahan, Palawan. Supreme Court of the Philippines, (August 16, 2023), <https://sc.judiciary.gov.ph/supreme-court-issues-writ-of-kalikasan-against-denr-and-mining-operators-in-mt-mantalingahan-palawan/>